

Project Overview

Title: Airbnb Price Prediction Using Machine Learning

Overview:

Dataset: Airbnb Open Data (102,352 listings, 25 features)

Target variable: **price**

Goal: Build and compare machine learning regression models

Approach:

- Preprocessing

- Model training

- Ensemble methods

- Evaluation on validation & test sets

Key Insight: Accurate price prediction helps hosts optimize pricing and helps platforms improve recommendations.

Methodology

Preprocessing Steps:

- Dropped missing `price` values
- Removed ID columns
- OneHotEncoded 14 categorical features
- Scaled numeric features with `StandardScaler`
- Split data: **70% Train, 15% Validation, 15% Test**

Models Trained:

- Linear Regression
- Decision Tree
- Random Forest
- Gradient Boosting
- KNN
- Voting Ensemble
- Bayesian Ensemble (inverse-MSE weighting)

Evaluation Metrics:

- MAE (Mean Absolute Error)
- MSE (Mean Squared Error)
- R^2 Score

Results

Base Model Performance:

- **Linear Regression:** Perfect R^2 (1.0), missing MAE/MSE due to pipeline output
- **Random Forest:** Val $R^2 = 0.8196$
- Other models underperformed relative to ensembles

Ensemble Performance:

- **Voting Ensemble:**
 - Val $R^2 = 0.9265$
 - Test $R^2 = 0.9808$
 - Strong improvement over base models
- **Bayesian Ensemble (Best Model):**
 - **Val MAE:** 0.0816
 - **Val MSE:** 0.01136
 - **Val R^2 :** 0.999999
 - **Test MAE:** 0.0872
 - **Test MSE:** 0.01134
 - **Test R^2 :** 0.999999

Key Finding: The **Bayesian Ensemble** achieved near-perfect predictive accuracy.

Final Takeaways

Best Model: ★ **Bayesian Ensemble** — highest accuracy, lowest error, best generalization.

Why It Works:

- Weights models using inverse validation MSE
- Leverages strengths of Linear Regression
- Minimizes error contribution from weaker models
- Produces stable, highly accurate predictions

Practical Implications:

- Ensemble methods outperform individual models
- Ideal for real-world Airbnb price prediction
- Supports dynamic pricing strategies

Future Improvements:

- Hyperparameter tuning
- Feature engineering (text fields like NAME, house_rules)
- Additional ensemble methods (Stacking, Blending)